

THE ± 2 -2 SURGERIES, ADJOINT ρ RHO-INVARIANTS, AND THE SHAPE OF A PROOF

Status of the claims. “Proof” means a complete argument from standard facts. “Computed” means checked by exact computation, by two independent programs, on all torus knots with $pq \leq 120$ (both chiralities) and on the figure-eight knot; these checks have not yet been refereed. “Conjecture” means what we expect but cannot yet prove.

1. The question

Let $K \subset S^3$ be a nontrivial knot and $Y_N = S_N^3(K)$. By Ni–Wu, Hanselman and Daemi–Lidman–Miller Eismeier (DLME), the cosmetic surgery conjecture reduces to showing that Y_2 and Y_{-2} are never orientation-preservingly diffeomorphic; one may assume $g(K) = 2$ and $\Delta_K = 1$, so in particular $\det K = |\Delta_K(-1)| = 1$. DLME settled the slopes $\pm 1/n$ with a real number $\ell(Y)$ extracted from Floer’s instanton homology and its Chern–Simons values, by proving a strict inequality between $\ell(Y_{1/n})$ and $\ell(Y_{-1/n})$. For ± 2 no such inequality is known. The aim of this note is to say precisely what ℓ is, where the constant $1/16$ comes from, and what we expect a proof to look like. The answer to the first two questions is: ℓ is an adjoint ρ -invariant, and $1/16$ is the ρ -invariant of one flat connection on the lens space $L(4, 1)$.

2. The invariant ℓ_{ell} is an adjoint ρ rho-invariant

Instanton homology. Let Y be a rational homology sphere with $H_1(Y) = \mathbb{Z}/2$, as for $Y_{\pm 2}$. There are two $\text{SO}(3)$ -bundles over Y : the trivial one P_0 and the one P_1 with $w_2 \neq 0$. For each, the irreducible flat connections (perturbed, if necessary, to be nondegenerate) generate a chain complex $C(Y, P_i)$, whose differential counts instantons on $\mathbb{R} \times Y$; its homology is $I(Y, P_i)$. A generator α carries two numbers: its Chern–Simons value $\text{CS}(\alpha) \in \mathbb{R}/\mathbb{Z}$, and its grading $\text{gr}(\alpha) \in \mathbb{Z}/8$, the index of the linearized instanton equation along a path from α to the trivial connection θ . A choice of path lifts these to $\widetilde{\text{CS}}(\alpha) \in \mathbb{R}$, $\widetilde{\text{gr}}(\alpha) \in \mathbb{Z}$; another path changes the pair by $(k, 8k)$, $k \in \mathbb{Z}$. Hence

$$c(\alpha) = \widetilde{\text{CS}}(\alpha) - \frac{1}{8} \widetilde{\text{gr}}(\alpha) \in \mathbb{R}$$

is well defined. DLME’s invariant is, in this language: for each degree d let $\kappa(d)$ be the least value of $\widetilde{\text{CS}}$ at which a nonzero homology class of degree d is represented, and $\ell(Y) = \inf_d (\kappa(d) - d/8)$.

The ρ -invariant. For a flat $\text{SO}(3)$ -connection α on Y , let $\text{ad } \alpha$ be the associated flat real vector bundle of rank three and

$$\rho(\alpha) = \eta(B_{\text{ad } \alpha}) - 3\eta(B),$$

where B is the odd signature operator of Y and $B_{\text{ad } \alpha}$ its twist by $\text{ad } \alpha$, and η is the Atiyah–Patodi–Singer spectral asymmetry. Thus $\rho(\theta) = 0$, ρ does not depend on the metric, and $\rho_{-Y}(\alpha) = -\rho_Y(\alpha)$. Write $h(\alpha) = \dim H^0(Y; \text{ad } \alpha)$: it is 3 for central connections, 1 for those with stabilizer a circle, and 0 for irreducibles.

Proposition 2.1 (proof: Atiyah–Patodi–Singer). *For every irreducible nondegenerate flat connection α on (Y, P_0) ,*

$$c(\alpha) = \frac{3 + \rho(\alpha)}{16}.$$

Proof. The lifted grading is the index of the linearized instanton operator on a path from α to θ , with small exponential decay imposed at θ . The index theorem of Atiyah–Patodi–Singer computes it as eight times the lifted Chern–Simons difference, corrected by half the spectral asymmetry of the operators at the two ends and half the dimension of the kernel at θ : $\widetilde{\text{gr}}(\alpha) = 8\widetilde{\text{CS}}(\alpha) - \frac{1}{2}(h(\theta) + \rho(\alpha) - \rho(\theta)) = 8\widetilde{\text{CS}}(\alpha) - \frac{3}{2} - \frac{1}{2}\rho(\alpha)$. Divide by 8. The signs are those fixed by DLME’s example: on the Poincaré sphere the two generators have $(\text{CS}, \text{gr}) = (1/120, 1), (49/120, 5)$, so $c = -7/60, -13/60$, and indeed $\rho = -73/15, -97/15$ there. $\square$

Definition 2.2. For a nonzero homology class x of $I(Y, P_i)$ let $\rho(x)$ be the least value, over cycles representing x , of the largest $\rho(\alpha)$ among the generators α occurring in the cycle. Put $\rho_*(Y, P_i) = \min_x \rho(x)$, the minimum over all nonzero classes, and

$$\lambda_i(Y) = \frac{3 + \rho_*(Y, P_i)}{16} \quad (i = 0, 1).$$

Corollary 2.3 (proof). $\ell(Y) = \lambda_0(Y)$. *In words: DLME’s invariant is the adjoint ρ -invariant of the lowest nonzero homology class, divided by 16 and shifted by $3/16$.*

Proof. Among generators of one lifted degree d , $\widetilde{\text{CS}} = c + d/8 = (3 + \rho)/16 + d/8$, so ordering by Chern–Simons value is ordering by ρ . Hence $\kappa(d) - d/8 = (3 + \min \rho(x))/16$, the minimum over classes of degree d ; take the infimum over d . $\square$

When the differential vanishes, as for every Seifert fibred Y , this reads $\lambda_i(Y) = \min_\alpha(3 + \rho(\alpha))/16$, and λ_i can be computed from the ρ -invariants of the flat connections alone. For P_1 there is no trivial connection to fix lifts, and we *define* λ_1 by the same formula; this is the normalization in which all maps below obey one rule, whatever their bundles. Since ρ is intrinsic, λ_0 and λ_1 are invariants of the oriented manifold, and an orientation-preserving diffeomorphism $Y_{-2} \rightarrow Y_2$ would give $\lambda_i(Y_{-2}) = \lambda_i(Y_2)$ for $i = 0, 1$ (it carries P_1 to P_1 , the only nontrivial bundle).

3. How cobordism maps move ρ

Let W be a cobordism from Y to Y' with $b_1(W) = 0$, carrying an $\mathrm{SO}(3)$ -bundle. Maps $I(Y) \rightarrow I(Y')$ are defined by counting instantons on W in zero-dimensional moduli spaces, possibly over a family of metrics of dimension g and cut down by conditions of total degree k ; put $i_0 = k - g$. Some constructions use a W with additional boundary components M_j (“middle ends”), on which the instantons are asymptotic to flat connections γ_j .

Proposition 3.1 (proof: Atiyah–Patodi–Singer). *If an instanton A of energy $E(A)$ on W , counted by such a map, runs from α on Y to β on Y' , then*

$$\frac{\rho(\beta) - \rho(\alpha)}{16} = \frac{i_0 + 3b^+(W)}{8} - E(A) - \sum_j \frac{3 - h(\gamma_j) + \rho(\gamma_j)}{16},$$

with each M_j oriented as an outgoing end.

Proof. The index theorem gives the index of A as $8E(A) - 3(1 + b^+(W))$ plus, for every end N with limit γ_N , the term $\frac{1}{2}(3 - h(\gamma_N) + \rho_N(\gamma_N))$. For the incoming end this is $\frac{1}{2}(3 - \rho(\alpha))$, for the outgoing end $\frac{1}{2}(3 + \rho(\beta))$. Set the index equal to i_0 and divide by 8. $\square$

So, in the units $\rho/16$, every map moves levels by $(i_0 + 3b^+)/8$ minus the energy, minus a term for each middle end. Write L for the largest value of the right side over the counted instantons and call it the *level* of the map. Exactly as in DLME:

Corollary 3.2 (proof). *If a map of level L is injective on homology, then $\lambda_i(Y') \leq \lambda_j(Y) + L$, for the bundles P_j on Y and P_i on Y' that the map uses.*

The first two terms lie in $\frac{1}{8}\mathbb{Z}$, and energies are positive. So *an odd multiple of $1/16$ can enter a level only through $\rho(\gamma_j)$ on a middle end.* Here are the values that occur (computed from the formula of Atiyah–Patodi–Singer for lens spaces, $\rho = \pm \frac{4}{p} \sum_{j=1}^{p-1} \cot \frac{\pi j}{p} \cot \frac{\pi q j}{p} \sin^2 \frac{\pi k j}{p}$ for $L(p, q)$ and the circle character k , the sign depending on orientation):

end	flat connection γ	$h(\gamma)$	$\rho(\gamma)$	$(3 - h + \rho)/16$
$S^3, L(2, 1), L(4, 1)$	central	3	0	0
$L(2, 1) = \mathbb{R}P^3$	rotation by π ($w_2 \neq 0$)	1	0	1/8
$L(4, 1)$	rotation by π ($w_2 = 0$)	1	± 2	1/4 or 0
$L(4, 1)$	rotation by $\pi/2$ ($w_2 \neq 0$)	1	± 1	3/16 or 1/16

The second row is DLME’s constant: their argument for ± 1 surgery uses a sphere of square -2 , whose neighbourhood has boundary $\mathbb{R}P^3$. The value $\rho = 0$ there is forced: $\mathbb{R}P^3$ has an orientation-reversing diffeomorphism, so $\rho = -\rho$. For ± 2 surgery the corresponding sphere S has square -4 (it is formed by the cores of the two 2-handles of $-X_2(K) \cup_{S^3} X_{-2}(K)$), and its neighbourhood has boundary $L(4, 1)$, which has no orientation-reversing diffeomorphism. So the last row splits the constant $1/8$ into $1/8 \pm 1/16$:

$\frac{1}{16} = \frac{1}{8} - \frac{|\rho(\gamma)|}{16}, \quad \gamma = \text{the flat connection of } L(4, 1) \text{ with holonomy a rotation by } \pi/2.$

Equivalently, $1/16$ is the energy of the least reducible instanton on the neighbourhood of S asymptotic to γ : for a sphere of square $-e$ this energy is $1/(4e)$, which is $1/8$ for $e = 2$ and $1/16$ for $e = 4$. Passing to the double cover branched along S turns $e = 4$ into $e = 2$ and doubles energies, which is why $1/16$ is “half of DLME’s $1/8$ ”.

Two consequences shape everything that follows. First, γ has $w_2 \neq 0$, so it occurs only for bundles on $-X_2(K) \cup X_{-2}(K)$ that are odd on S ; such a bundle is trivial on one of $Y_{\pm 2}$ and nontrivial on the other. *The maps that carry the constant $1/16$ always exchange P_0 and P_1 .* Second, nothing forces a gap between $\lambda_0(Y_2)$ and $\lambda_0(Y_{-2})$ alone.

4. What the examples say

All quantities below are computed, in the normalization of Definition 2.2. Put

$$m_1 = \lambda_0(Y_2) - \lambda_1(Y_{-2}), \quad m_2 = \lambda_1(Y_2) - \lambda_0(Y_{-2}).$$

	$\lambda_0(Y_2) - \lambda_0(Y_{-2})$	$\lambda_1(Y_2) - \lambda_1(Y_{-2})$	m_1, m_2
positive $T(p, q)$, $n = pq$	$-\frac{1}{8} + \frac{2n-5}{n^2-4}$ (changes sign)	$\frac{3}{8} - \frac{4}{n^2-4}$	in $(\frac{1}{8}, \frac{3}{16}]$
negative $T(p, q)$	in $[\frac{3}{32}, \frac{1}{8}]$	in $[0.085, \frac{1}{8}]$	in $[\frac{1}{16}, \frac{5}{32}]$
figure-eight knot	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{13}{80}, \frac{1}{16}$

So the same-bundle difference is not bounded below by $1/16$ (it fails for the eleven positive torus knots with $10 \leq pq \leq 29$), while the mixed differences m_1, m_2 lie in $[\frac{1}{16}, \frac{3}{16}]$ in every case, with $1/16$ attained (figure-eight knot, mirror trefoil). The minimum over both bundles, $\min(\lambda_0, \lambda_1)(Y_2) - \min(\lambda_0, \lambda_1)(Y_{-2})$, is at least $\frac{1}{8} - \frac{1}{884}$ in every case. A cosmetic pair would make all of these differences zero.

5. The expected theorem and the shape of its proof

Conjecture 5.1. *For every nontrivial knot K , $m_1 \geq \frac{1}{16}$ and $m_2 \geq \frac{1}{16}$; if $\det K = 1$, both inequalities are strict.*

Corollary 5.2 (proof, given the conjecture). *$S_2^3(K)$ and $S_{-2}^3(K)$ are not orientation-preservingly diffeomorphic.*

Proof. A diffeomorphism gives $\lambda_i(Y_{-2}) = \lambda_i(Y_2)$, so $m_1 + m_2 = 0$, while the conjecture gives $m_1 + m_2 \geq \frac{1}{8}$. $\square$

We expect the proof to have four steps.

Step 1: two exact triangles that exchange bundles (conjecture). Let $T(K)$ be the instanton homology of S^3 relative to K whose generators are the representations of the knot group sending the meridian to a trace-zero element of $SU(2)$ (Kronheimer–Mrowka’s singular instanton homology). We expect long exact sequences

$$\cdots \rightarrow I(Y_2, P_0) \xrightarrow{a} T(K) \xrightarrow{b} I(Y_{-2}, P_1) \xrightarrow{c} I(Y_2, P_0) \rightarrow \cdots,$$

and the same with P_0, P_1 exchanged. Here a and b come from the traces of the two surgeries, with the cores of the 2-handles as singular surfaces; c comes from a cobordism from Y_{-2} to Y_2 with one positive direction and one further boundary component $L(4, 1)$, along which the instantons are asymptotic to the connection γ of the table. By Proposition 3.1 the levels are: a, b at most 0; c equal to $\frac{3}{8} - \frac{3}{16} = \frac{3}{16}$ (one positive direction, and the γ -term); the null-homotopy of $b \circ a$, a count over a one-parameter family of metrics, at most $-\frac{1}{8} - \eta$ with $\eta > 0$. These are the two constructions that each failed on its own (the family on odd bundles, and the family singular along S): the conjecture is that they are the two halves of one triangle, with their $L(4, 1)$ -faces cancelling. Evidence (computed): the ranks satisfy $\text{rk } I(Y_{-2}, P_1) = \text{rk } I(Y_2, P_0) + \#\{\text{traceless representations}\}$ for 29 torus knots of both chiralities, maps of the predicted levels exist in all 58 cases, and the same method reproduces DLME’s ± 1 inequality in 25 of 25 cases.

Step 2: filtered algebra (proof, given Step 1). Assume the differentials vanish, as in all examples. Let x be a generator realizing $\lambda_0(Y_2)$. Either $a(x) = 0$, or $a(x) \neq 0$ in $T(K)$. In the first case, exactness and the level of the null-homotopy give $\lambda_1(Y_{-2}) \leq \lambda_0(Y_2) - \frac{1}{8} - \eta$, that is $m_1 > \frac{1}{8}$; in the second, $T(K)$ has a class of level at most $\lambda_0(Y_2) - \eta_a$, where η_a is the least energy counted by a . The same holds for m_2 . (This is DLME’s argument, with the vanishing group $I(S^3) = 0$ replaced by $T(K)$.)

Step 3: the cosmetic case (proof, given Step 1). If $Y_{-2} \cong Y_2$ and no irreducible traceless representation sends the longitude to ± 1 , Step 2 run around both triangles shows that a is nonzero on the lowest classes of both bundles, and that

$$\lambda(T(K)) < \min(\lambda_0, \lambda_1)(Y_2) \leq \max(\lambda_0, \lambda_1)(Y_2) < \lambda(T(K)) + \frac{1}{16},$$

where $\lambda(T(K))$ is the lowest level of $T(K)$, defined as in Definition 2.2. So a cosmetic pair needs both lowest classes to map into $T(K)$ at a cost in energy below $1/16$.

Step 4: the missing estimate (conjecture). The map a costs energy more than $1/16$ on the lowest class. The examples locate the difficulty: the two cases with $m_2 = 1/16$ (figure-eight knot, mirror trefoil) are exactly those in which the lowest class maps to $T(K)$ at zero energy, through a binary dihedral representation; such representations do not exist when $\det K = 1$. Locally, along the curve of representations of the knot group, the estimate holds where the curve crosses the trace-zero line with negative slope and can fail where it crosses with positive slope; a proof must use a global property of that curve when $\det K = 1$.

What remains. (i) The triangles of Step 1: gluing along $L(4, 1)$ at γ with a singular cap, exclusion of breaking at the reducible of (S^3, K) , and the degree argument of Culler–Daemi–Xie. (ii) The estimate of Step 4. (iii) When the differentials do not vanish, control of how far below the lowest class its boundaries can reach. Steps 2 and 3 are formal. The analysis in (i) is of the same kind as in DLME and in Kronheimer–Mrowka’s singular theory; no $SO(3)$ -monopoles and no families beyond one parameter are needed. If (i) and (ii) hold, the remaining argument is comparable in length to DLME’s.